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Minireview

Immunogenicity of Cas9 Protein



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Clustered regularly interspaced short palindromic repeats (CRISPR) form the adaptive immune system in archaea and bacteria and have been modified for genome engineering in eukaryotic cells. CRISPR systems contain 2 components, a single-guide RNA, which is a short RNA composed of a 20 nucleotide sequence that targets specific sites in the genomic DNA and a scaffold necessary for its binding to the CRISPR-associated endonuclease (Cas9). Because of its high efficiency and accuracy, the CRISPR-Cas9 genome editing based therapies are poised to treat a multitude of human diseases with a promise to target previously “undruggable” proteins. As the first in-body clinical trial with

CRISPR-Cas9 is embarked on, the risks associated with administering the genome editing machinery to patients become increasingly relevant. Recent studies have demonstrated an innate and adaptive cellular immune response to Cas9 in mouse models and the presence of anti-Cas9 antibodies and T-cells in human plasma. Pre-existing immunity against therapeutic Cas9 delivery could decrease its efficacy in vivo and may pose significant safety issues. This review focuses on the immunogenicity of the Cas9 protein and summarizes potential approaches to circumvent the problem of immune recognition.

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Introduction

The CRISPR/Cas9 system borrows a microbial adaptive immune defense system as a promising approach to carry out targeted genetic changes in eukaryotic cells. Clustered regularly interspaced short palindromic repeats (CRISPR) associated nuclease 9 (CRISPR-Cas9) has lately attracted a lot of interest as an RNA-guided genome editing tool which comprises a nuclease, Cas9, and a single-guide RNA (sgRNA) that recognizes target DNA and guides the Cas9 protein to the target loci adjacent to a protospacer adjacent motif and generates site-specific double-strand breaks that are subsequently repaired either by nonhomologous end-joining, which is more efficient, or by the more precise homology-directed repair on the existence of a donor template.^{1,2} While it was not the first genome editing strategy available, CRISPR-Cas9 has proven to be a powerful tool for this purpose because of its ease of use, site-specific activity, and limited off-target effects.² The potential applications in medicine are ample, ranging from disrupting mutant alleles in cancer³ or repairing mutated alleles causing monogenic disorders, such as muscular dystrophy⁴ or sickle cell anemia.⁵ Current preclinical cell therapy pipelines are filled with CRISPR-Cas9 genome editing

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strategies, either through ex vivo editing of cells followed by their transplantation to the patient or in vivo editing of patients' cells; however, concerns regarding its safety and efficacy remain. Prior gene transfer studies have shown strong host immune responses, thereby limiting their therapeutic advantage. Gene editing runs similar risks concerning immune recognition, first from the delivery vector, followed by the genome editing components, Cas9 and sgRNA, and finally the expression of the edited gene. For this review, we will focus on the host immune response to the nuclease, Cas9, as immunogenicity of gene delivery vectors have extensively been reviewed elsewhere.^{6,7}

The 2 most commonly used Cas9 orthologues are derived from *Staphylococcus aureus* (SaCas9) or *Streptococcus pyogenes* (SpCas9), both of which are prevalent human commensals that could be pathogenic, for instance causing strep throat. Approximately, 40% of the human population is colonized by *S. aureus*⁸ and 12% of the children under 18 years have an asymptomatic colonization with *S. pyogenes*.⁹ Notably, both humoral, antibody-mediated, and cellular, T cell-mediated immunity have been detected against *S. aureus* and *S. pyogenes* in 80% of healthy individuals.^{10,11} However, most of these responses are against secreted proteins and proteins found on the membrane surface of the bacteria, which are easily accessible to the immune system. Because Cas9 is an intracellular protein and most therapeutic interventions aim to temporarily express or deliver the recombinant Cas9 directly to target cells, it can

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be hypothesized that anti-Cas9 antibodies would be negligible.¹² Contrary to this assumption, Wang et al. in their seminal study observed SpCas9 specific antibodies 14 days after adenoviral Cas9 delivery. They aimed to disrupt Pten expression in hepatocytes, thereby developing a mouse model for human nonalcoholic steatohepatitis.¹³ While they successfully achieved genome editing, they observed IgG1, IgG2a, and IgG2b subtypes of antibodies, indicating a host immune response to adenoviral Cas9. Furthermore, Chew et al.¹⁴ demonstrated that regardless of the delivery method, expression of Cas9 in mice evoked an immune response (Fig. 1). Cas9 expressing muscles showed a consequent enrichment in CD45⁺ leukocytes, especially myeloid cells (CD11b⁺Gr1⁺ monocytes, macrophages and dendritic cells subsets) and T cells (CD3⁺CD4⁺ and CD3⁺CD8⁺). Interestingly, they identified 4 T cell receptor β -chain clonotypes that were common to all Cas9 exposed animals, of which one clonotype was common among all animals and recognized as true Cas9-responsive (Fig. 1, right).

Chew et al. also observed varying titers of Cas9-specific antibodies in individual mice after Cas9 exposure, indicating a largely individual humoral response. Cas9 epitope mapping revealed that 3 linear epitopes appeared repeatedly. These 3 residues are essential for Cas9 function, and include the gRNA recognizing residues, the REC-1 domain which contributes to Cas9-gRNA interactions and the protospacer adjacent motif binding loop. Taken together, these results demonstrate that CRISPR-Cas9 evokes a host cellular and humoral immune response with distinct cellular and molecular signatures, indicating that Cas9 can serve as an antigen in mammals.

More recently, by probing human serum for the presence of anti-Cas9 antibodies using an enzyme-linked immunosorbent assay, Charlesworth et al.¹⁵ detected immunoglobulin G (IgG) antibodies against both SaCas9 and SpCas9 in 78% and 58% of the donors, respectively. They next evaluated the presence of antigen reactive T cells, indicative of a pre-existing immune response to Cas9. Using 3 different highly sensitive assays, they detected a high frequency of cytokine-positive antigen-reactive T-cells against SaCas9 (78% donors) and SpCas9 (67% donors). They found that all donors positive for cellular activity against Cas9 were also positive for antibody activity, demonstrating a high concordance between adaptive and humoral immunity. These results indicate that humans are often exposed to and mount an immune response to Sa- and SpCas9, wherein subsequent exposure to the protein would result in antibodies that bind to and render the protein inactive or form immune complexes that are actively cleared by the circulation. A potential option would be to screen patients for anti-Cas9 antibodies and T-cells, before any gene editing based

therapeutic interventions; however, the large proportion of the population to which this applies prompts its reassessment. Another point of concern would be to patients receiving Cas9-based therapies in the hospitals because there is an increased risk of *Staphylococcus* infections in hospitals^{5,16} and any simultaneous *S. aureus* or *S. pyogenes* infection may activate dormant anti-Cas9 cytotoxic T lymphocytes (CTLs) rendering the treatment ineffective.

Recently, 2 other groups have reported pre-existing adaptive immune responses to Cas9 in humans. Simhadri et al.¹⁷ detected anti-SaCas9 antibodies in 10% and anti-SpCas9 antibodies in 2.5% of the tested population (200 donors). Notably, Simhadri et al.¹⁷ report a considerably lower prevalence of anti-Cas9 antibodies as measured by ELISA as Charlesworth et al.,¹⁵ who analyzed anti-Cas9 antibodies by immunoblotting. In addition, the sample size in both reports is markedly different, with 48 serum samples in the training set and 200 samples used in the validation set of Simhadri et al.¹⁷ as compared to 22 cord blood and 12 serum samples analyzed by Charlesworth et al.¹⁵ ELISA-based assays are more sensitive, quantitatively measure unaltered epitopes, allowing for setting statistically determined cutoff points. On the other hand, immunoblotting measures denatured proteins and depends on the sensitivity of the antibodies, on detection reagents used, and on exposure times, resulting in a more qualitative and subjective analysis of the bands. Taking into consideration the differences observed in the 2 reports, and the impact of anti-Cas9 antibodies on the further development of Cas9-based therapies, standardized analysis methods to resolve these differences would be critical. Wagner et al.¹⁸ demonstrated that SpCas9 challenge to peripheral blood mononuclear cells could activate T effector (T_{eff}) cells, indicated by CD137 upregulation in both CD4⁺ and CD8⁺ T cell compartments. Interestingly, SpCas9 induced an effector-memory (T_{em}) T cell response accompanied by an increase in regulatory T cells (T_{reg}) in humans. Furthermore, in vitro studies demonstrated that the endogenous SpCas9-reactive T_{reg} cells have the potential to alleviate the activation, expansion, and function of SpCas9-reactive

T_{eff} cells. Therefore, alterations in T_{reg}/T_{eff} cells' ratios and strong CD8⁺ T cell responses could pose potential limitations on Cas9-associated therapies, while providing a possible strategy for modulating the impact of the pre-existing immunity to Cas9 in the future. Taking these results together with those of Charlesworth and Simhadri et al., additional studies monitoring a Cas9-reactive immune response before and during Cas9-based therapeutic interventions (in clinical trials) would be essential to potentially identify high-risk patients with misbalanced T_{reg}/T_{eff} ratios and high CD8⁺ T cell responses.

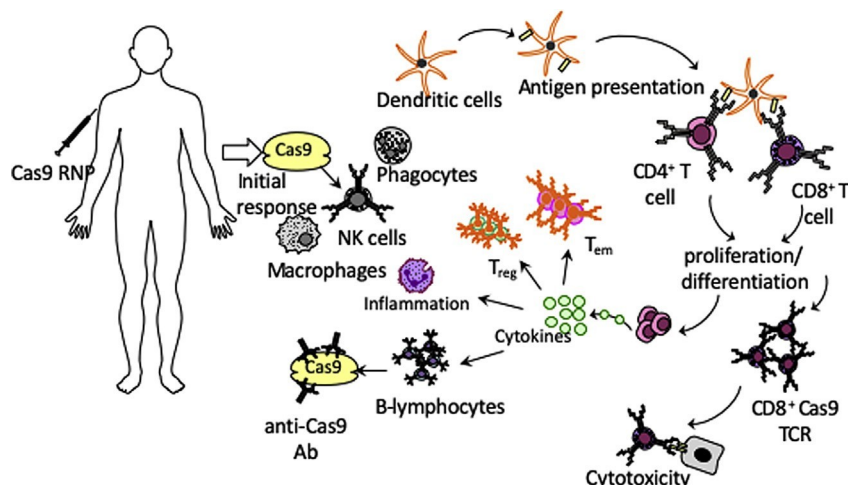


Figure 1. The immune response to Cas9. Injection of CRISPR machinery triggers an innate and adaptive immune response. 64
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While the pre-existence of an adaptive immune response to Cas9 may not prove to be a major barrier for the implementation of ex vivo therapies wherein cells are treated in the absence of an immune system and implanted after the complete degradation of the Cas9 protein, it poses a significant threat to the development of in vivo therapies. When patients with pre-existing antigen-reactive T cells against Cas9 encounter recombinant Cas9 delivered in a therapeutic context, their memory T cells might rapidly expand in response to Cas9 being presented on the cell surface by major histocompatibility complex (MHC) class I, resulting in recognition and targeting by cytotoxic T cells and ultimately clearing the cells expressing Cas9, rendering the therapy ineffective. Notably, while a single injection of Cas9-ribonucleoprotein (RNP) may be tolerated by patients without pre-existing anti-Cas9 immunity, a significant immune response may occur after first exposure. After an immune response is generated, any subsequent treatments using Cas9-RNPs would be limited, which may be critical for the treatment of certain diseases. To circumvent these problems, it is inessential to develop strategies whereby a Cas9-directed immune response can be avoided.

Strategies to Evade Pre-Existing Immunity to Cas9

To minimize the chance of an adverse immune response after CRISPR-Cas9 gene therapy, known strategies may be used, such as structural modification of Cas9 proteins thereby masking immunogenic epitopes, using Cas9 orthologs from nonpathogenic bacteria, inducing immune tolerance or immune suppression or to target immune privileged organs, such as the eye.

Masking Immunogenic Cas9 Epitopes

A potential solution to evade an immune response would be "epitope masking," wherein potentially immunogenic peptide sequences are identified and modified or removed to prevent their detection by the immune system, while still maintaining the function of the original protein. In a remarkable study published earlier this year, Ferdosi et al.¹⁹ identified 2 immunodominant SpCas9 T cell epitopes for the human leukocyte antigen (HLA)-A*02:01 (Fig. 2) using an enhanced prediction algorithm that incorporates T cell receptor contact residue hydrophobicity and HLA binding and evaluated them by T cell assays using healthy donor peripheral blood mononuclear cells. Furthermore, they eliminated the

immunodominant epitopes after targeted mutation of the Cas9 protein, while preserving its function and specificity. Two mutations in the anchor residues of the Cas9 epitope were explored individually and in combination to assess their effect on immunogenicity. Single amino acid mutations in this region could produce an "incognito" version of Cas9. In vitro, they demonstrated that the mutated Cas9 retained its nuclease activity intact while T-cell reactivity to the mutated peptide showed a 25- to 30-fold reduction.

Altering Antigen Presentation of Cas9 Epitopes

Another promising approach derives from Epstein Barr virus (EBV) which interferes with proteasomal degradation and subsequent antigen presentation.²⁰ The EBV-encoded nuclear antigen (EBNA1), expressed in latent EBV-infected B lymphocytes persists in carriers and is the only viral protein detected in all EBV-associated malignancies. Interestingly, however, MHC class I-restricted, EBNA1-specific CTL responses have not been demonstrated. It was shown that repetitive glycine-alanine (Gly-Ala) generate a cis-acting inhibitory signal that interferes with antigen processing and MHC class I-restricted presentation, a mechanism of viral escape from CTL surveillance which promotes viral persistence. Accordingly, introducing such Gly-Ala repeats into Cas9 might similarly restrict proteasomal degradation of Cas9 and presentation to CTLs via HLA class I.²¹ However, it remains to be seen if the functionality and specificity of the protein can be maintained after changing the protein sequence.

While immunoproteasome inhibition using small molecule inhibitors such as ONX 0914,²² the FDA-approved inhibitors bortezomib (Velcade) or carfilzomib (Kymrius)²³ have been successfully used in hematological malignancies inflammatory and autoimmune diseases, they target the catalytic activity of both, the constitutive and immunoproteasome indiscriminately. Such inhibitory effects on the constitutive proteasome are expected to result in severe unwanted side effects, which would not be a solution to the immunogenicity dilemma.

Cas9 Orthologs From Nonpathogenic Bacteria

Over the last decade, several CRISPR-Cas9 orthologs have been discovered,²⁴ which could potentially be tested for their efficacy and immune recognition. CRISPR-Cas9 systems from nonpathogenic bacteria or bacteria from extreme habitats

without any prior exposure to humans could offer a potential solution, circumventing

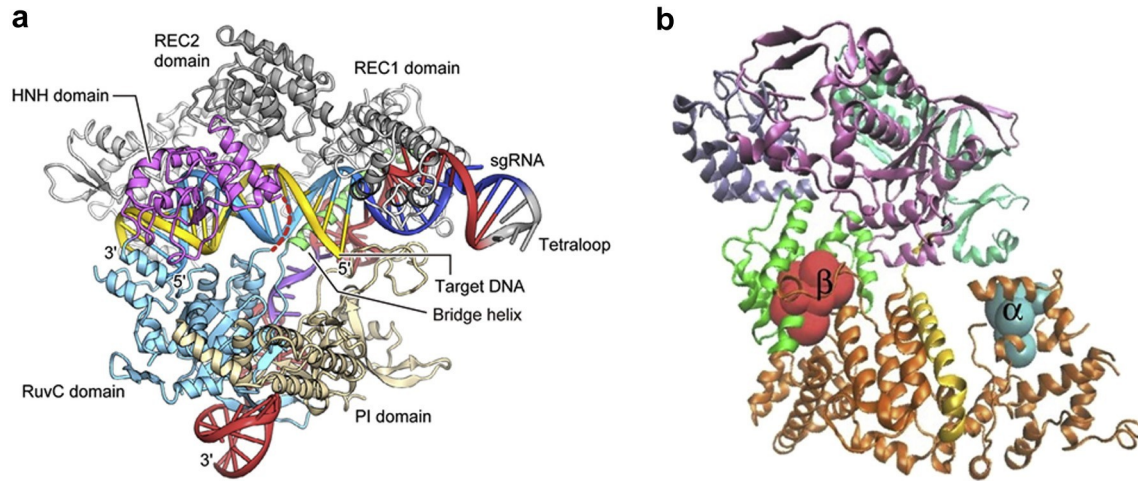


Figure 2. (a) Overall structure of the Cas9-sgRNA-DNA ternary complex. Ribbon representation of the Cas9-sgRNA-DNA complex. Disordered linkers are shown as red dotted lines. Reproduced with permission from ³⁶. (b) 3-D structure of the SpCas9 protein, showing the location of the identified immunodominant epitopes α and β . Reproduced with permission from ¹⁹.

Table 1 Clinical Trials Involving CRISPR/Cas9

Type of Disease	Target Gene	Description	CRISPR Strategy	Delivery Route	Status	Sponsor or Affiliations	NCT ID
β -thalassemia, sickle cell anemia	BCL11A	Ex vivo ablation of BCL11A by CRISPR-Cas9 in autologous CD34 β hematopoietic stem cells	NHEJ	Electroporation of cells with sgRNA and SpCas9. Cells infused via central venous catheter.	Phase 1/2	CRISPR Therapeutics, Vertex Pharmaceuticals	NCT03655678
B cell leukemia and lymphoma	e	Ex vivo modified universal CRISPR-Cas9 gene-editing CAR-T cells targeting CD19 (UCART019) in patients with relapsed or refractory CD19 β leukemia and lymphoma	e	Intravenous injection of modified cells	Phase 1/2	Chinese PLA General Hospital	NCT03166878
Metastatic nonesmall cell lung cancer	PD-1	Dose-escalation study of ex-vivo knocked out, expanded, and selected PD-1 knockout-T cells from autologous origin	NHEJ	Intravenous infusion of modified cells	Phase 1/2	Sichuan University	NCT02793856
HIV-1-infection	CCR5	CD34 β hematopoietic stem/progenitor cells from donor are treated with CRISPR/Cas9 targeting CCR5.	NHEJ	Not indicated	Phase 1/2	Affiliated Hospital to Academy of Military Medical Sciences, Peking University, Capital Medical University	NCT03164135
Leber congenital amaurosis type 10	CEP290	In vivo CRISPR-Cas9 gene editing repair correcting the splicing defect in CEP290 caused by the mutation, IVS26	NHEJ	Single dose subretinal injection of AAV5 vector encoding sgRNA and SaCas9	Phase 1/2	Editas Medicine, Allergan Pharmaceuticals	NCT03872479
Hereditary transthyretin amyloidosis	TTR	In vivo CRISPR/Cas9 gene editing to disrupt mutant TTR allele	NHEJ	In vivo lipid nanoparticle containing sgRNA and SpCas9 mRNA	Filing planned (2019)	Intellia Therapeutics, Regeneron Pharmaceuticals	Not indicated

NHEJ, nonhomologous end-joining.

the problem of pre-existing antibodies and antigen-specific T cells. For instance, Cas9 from *Geobacillus stearothermophilus*, inhabiting soil, thermal vents, and oceanic sediment was found to be highly stable in human plasma²⁵ when compared to SpCas9. Using GsCas9 for instance for direct (i.v.) injection of Cas9-sgRNA RNPs could significantly improve genome editing efficiencies. While, in principle, GsCas9 might be just as immunogenic as SpCas9 and illicit an immune response in vivo, it allows for one Cas9-sgRNA exposure before the onset of any immunity. Consequently, using GsCas9 coupled to a cautious dose of immunosuppression, as applied in transplant patients, would be beneficial.

Immune Privileged Organs

To lessen the risk of an immune response to Cas9, immune privilege may be exploited, limiting the body's natural immune response and providing a microenvironment actively accommodating of foreign proteins. The eye is one of a few areas of the body with immune privilege. The eye limits its inflammatory immune response so that vision is not harmed by swelling and other tissue changes. A remarkable example is the first in human clinical trial for gene editing therapy where subretinal injection of Cas9-encoding DNA and 2 sgRNAs is tested as a therapeutic intervention for Leber's congenital amaurosis type 10, the most common form of inherited childhood blindness.²⁶ The association between an immune response and intraocular inflammation associated with intravitreal injection of recombinant adeno associated virus (AAV) 2 was examined in 15 patients and monitored for up to 96 weeks after injection.²⁷ Ocular injection of AAV2 in these patients appeared to be well tolerated, with a mild increase in serum AAV2-neutralizing antibody titers. Most patients, however, did demonstrate a mostly mild, self-limited intraocular inflammation response to the injection which was resolved with topical anti-inflammatory agents or oral corticosteroids, without any long-term sequelae, supporting the continued use of gene therapy-based approaches in eye diseases. However, it remains to be seen if antibodies against AAV2 can be detected in the eye after injection, indicative of a local immune response to gene therapy. Other sites with immune privilege include the brain, testes, placenta, and fetus. In utero genome editing could potentially treat genetic diseases prenatally that, if left untreated, would result in significant morbidity and mortality before or shortly after birth. In an initial proof of concept study, Rossidis et al.²⁸ used AAV-mediated delivery of Cas9 and sgRNA targeting Hpd via vitelline vein injection in a mouse model of hereditary tyrosinemia type 1 (HT1). In wild-type mice, they achieved approximately 15% genome editing in the liver at 2 weeks age, whereas all other organs showed no genome editing. In HT1 mice, which experience neonatal lethality, over 35% genome editing was observed in the liver at 1 and 3 months of age. While these results are initial stepping stones to therapy for selected congenital genetic disorders, the route to human application is still long and full of ethical and safety concerns.²⁹ Corners regarding in

utero interventions are amplified, such that the risk of infection, immune reactions, and the induction of preterm labor are faced by 2 parties, the mother and the fetus. Moreover, because the delivery of the gene editing machinery cannot be adequately controlled in utero, its effects cannot be assessed until birth and an additional concern that the permanent changes may be made to the germline of the developing fetus, which can be passed onto subsequent generations.²⁸

Induction of Immune Tolerance

As seen by Wagner et al., the expansion of T_{reg} cells after SpCas9 challenge restricted the proliferative response of Cas9-specific T_{eff} cells without altering unrelated antigen-specific T_{eff} cells. Because a large proportion of the population shows the prevalence of T_{reg} cells within circulation, a potential strategy would be to expand this natural resource within the body to induce an immune tolerance to the Cas9 protein. Following ex vivo expansion of Cas9-specific T_{reg} cells from patients and their reinjection before gene editing therapies might be able to minimize or control an immune response but seems to be the most invasive strategy of overcoming critical hurdles.

Furthermore, the liver has naturally acquired a specialized immune tolerance, which is indispensable to the body, avoiding any overactivation of the adaptive and innate immune responses toward nutrients, gut-derived bacterial metabolites and lipopolysaccharides, and cellular debris that enter the liver via the portal vein from the intestine.³⁰ This liver tolerance effect was first described in 1969 in pigs wherein liver allografts were accepted without immunosuppression despite an MHC complex mismatch.³¹ Keeping this in mind, the liver presents itself as a preferred target organ for gene therapy for liver-specific disorders but also for systemic delivery of proteins. Over the last decade, several studies have demonstrated tolerance to in vivo gene transfer and transgene expression in animals.³²⁻³⁴ Hepatic gene transfer offers a solution to 2 challenges, that is, first efficient gene transfer allows for therapeutic levels of the transgene to be expressed in vivo, and the hepatocyte-derived expression of the protein could tolerate the immune system to the foreign protein.

Conclusions

Heralded by science as the breakthrough of the year 2015,³⁵ CRISPR-Cas9-mediated genome editing offers unparalleled opportunities for therapy of monogenic diseases and complex, multigenic diseases like cancer. CRISPR-Cas9 offers multiple advantages to its counterparts such as zinc finger nucleases, including its ease of use, inexpensive workflow, and high specificity and flexibility. Its tremendous potential can be observed in the short time of less than a decade between its discovery to its first clinical use (Table 1). While its ex vivo use and application is being celebrated, the delivery of Cas9 and sgRNA directly to the cell of interest/diseased cell would definitely be a more

effective therapeutic strategy. The recent demonstration that routine prior exposure to *S. aureus* and *S. pyogenes* has resulted in pre-existing immunity to Cas9 poses a considerable threat to its further clinical development. However, initial studies using immune privileged organs, immunosuppression or previously un-encountered sources of Cas9 provide opportunities to circumvent these initial roadblocks to gene editing-based therapeutics.

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